System Analysis & Design

Introduction to UML

Business Use case

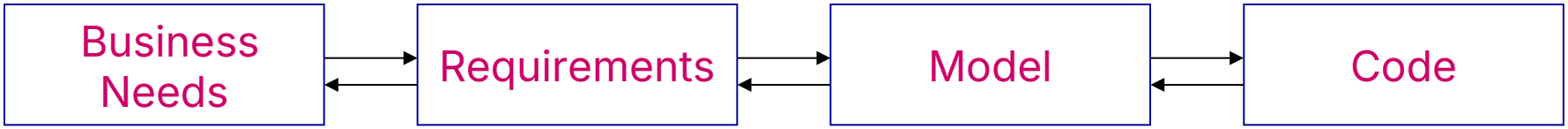
System Development Life Cycle (SDLC)

- Requirements Collection
- Requirements Analysis
- System Design
- System Development
- System Testing
- System Deployment
- System Maintenance

?System Modeling – How

- A model helps you *describe and plan* a system before you build it
  - Take the *business needs* of the users
  - Map them into *requirements* that your team can use and understand
  - Put these requirements into *a model* and generate *code* from it

By mapping the requirements to the code via the model, you can ensure that the requirements are actually met by the code, and that the code can easily be traced back to the requirements



Visual Modeling

- Visual Modeling is the process of taking the information from the model and displaying it *graphically* using a standard set of graphical elements

- A *standard* is vital to realizing one of the benefits of visual modeling: *communication*

- Communication between
  users, developers, analysts, testers, managers, and anyone else involved with a project

?What is UML

- The *Unified Modelling Language (UML)* is a standard language for specifying, visualizing, constructing, and documenting the software systems

- The UML is a very important part of developing *object oriented software*

- The UML uses mostly *graphical notations* to express the design of software projects

Benefits of using UML

- Help project teams *communicate*
- *Explore potential designs*
- *Validate* the architectural design of the software
- Be *independent of particular programming* languages and development processes

UML - Unified Modeling Language

UML DIAGRAMS

1. Business Use Case diagram
2. Use Case diagram
3. Activity diagram
4. Sequence diagram
5. Collaboration diagram
6. Class diagram
7. Statechart diagram
8. Component diagram
9. Deployment diagram

These model diagrams illustrate *different aspects of the system*

Business Use Case Diagram -1

- Business Use Case diagram is used to represent the *functionality provided by an organization as a whole*

- They answer the questions *"What does the business do?"*

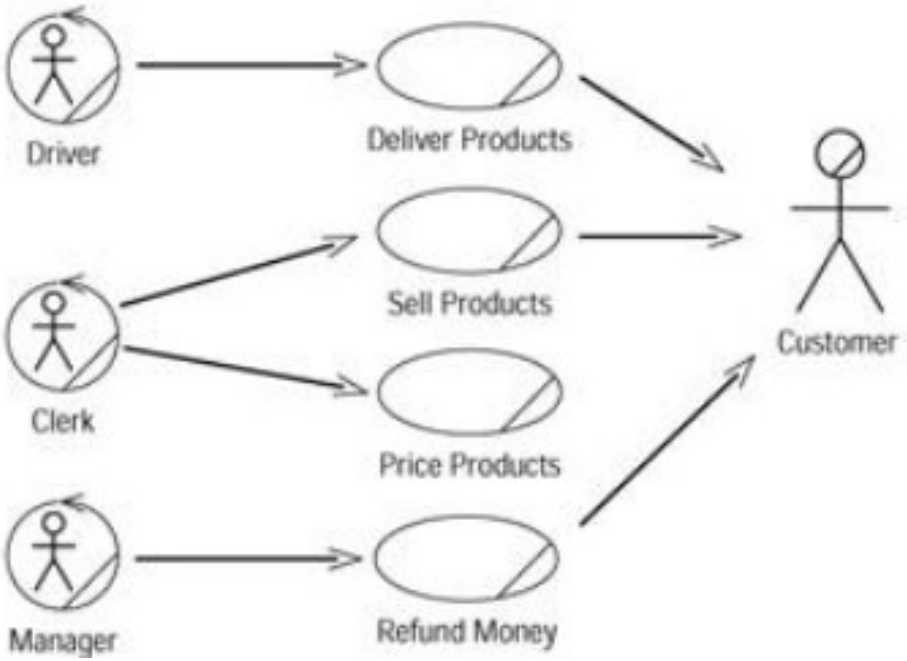
- Business Use Case diagram is drawn from the ***organizational perspective***

Elements of Business Use Cases

Business Use Case diagram shows the *interactions between business use cases and business actors & workers*

Element	Representation	Notation
Business actors	A business actor is anyone or anything that is *external* to the organization but interacts with it (Individual, group, company,...) *Secondary Actor*	
Business use cases	It represents the workflow within the organization. It keeps focus on what the business is *doing* Named in the form of *<verb><noun>*	
Business workers	A business worker is a role within the organization *Primary Actor*	
Associations	An arrow between a business actor or a business worker and a use case suggests that the actor or worker interacts with the use case	

Example


Driver
Deliver Products
Clerk
Sell Products
Price Products
Manager
Refund Money
Customer

Business Use Case Categories

Three categories of business use cases:

1.Business Processes

- The activities that involve *business actor interaction*

2.Supporting Processes

- The activities that *have to be performed anyhow to make the business work.* For example, systems administration, cleaning and security

3.Management Processes

- The activities that show the type of work that affects *how the business is managed* and *the relationships between the manager, and people who work in the business*

Why use Business Use Case ?Diagram

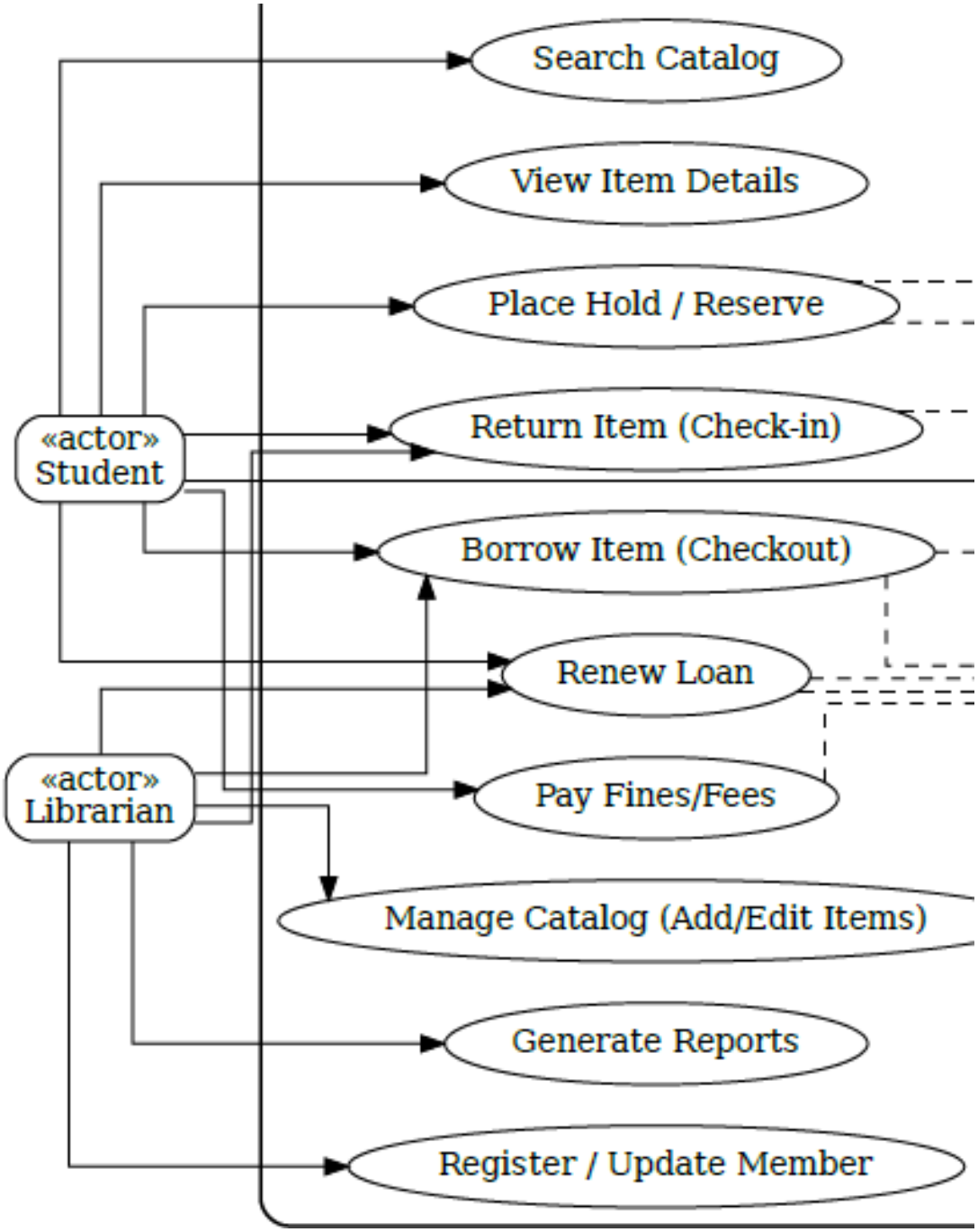
- When using UML to build software, business modeling can help to *understand* the *context of the system to be build*

- If we fail to understand the business, we may make *faulty assumptions* about what the software should do

- Business modeling gives the team a view of *the business itself* and *the workflows within it*

Practice

- Draw a Business Use Case Diagram for a library


Search Catalog
View Item Details
Place Hold / Reserve
Return Item (Check-in)
Borrow Item (Checkout)
Renew Loan
Pay Fines/Fees
Manage Catalog (Add/Edit Items)
Generate Reports
Register / Update Member
«actor»
Student
«actor»
Librarian

Thank You ☺